

Elective III

UEI735: POWER ELECTRONICS FOR AUTOMATION

L T P Cr.

3 0 2 4.0

Course objective: To review the operational aspects of power electronic devices and principle of conversion and control of AC and DC voltages, concept of rotating machines, and drives.

Introduction: Introduction to Thyristors and its family, turn-on and turn - off methods, IGBT, selection of devices for various applications.

Phase Controlled Converters: Principle of phase control, Single phase and three phase converter circuits with different types of loads.

AC Voltage Controllers: Types of single-phase voltage controllers, single-phase voltage controller with R and RL type of loads for industrial applications.

Inverters: Single phase voltage source bridge inverters, single-phase PWM inverters, current source inverters, types of inverters and their applications, Three phase bridge inverters with 180° and 120° modes.

Electrical Machines Control: Principle and characteristics of DC machines, speed control of DC motor, Principle of operation of single phase and three phase Induction motor, current-limit control.

DC Motor Drives: Principle of chopper operation, step-up and step-down choppers, EMF and torque production of DC motor, four quadrant operation, thyristor and chopper fed dc motor drives and its industrial applications.

Induction Motor Drives: Principles of operation, single phase to single phase step-up and step-down cycloconverters, Torque-speed and torque-slip characteristics, methods of starting of squirrel cage motors, generating and braking modes, variable frequency operation and its applications.

Laboratory Work: V-I characteristics of SCR, IGBT, DIAC and TRIAC, Gate firing circuit, DC -DC chopper, Semi converter and Full converter with R, RL and RLE type of loads, DC shunt motor speed control, Simulation of power electronics converters.

Course learning Outcomes (CLO):

After the completion of the course the students will be able to:

- 1 Exhibit the knowledge of various power semiconductor devices related to its characteristics, and to select semiconductor devices for various applications
- 2 Exhibit the knowledge of working principle and applications of phase controlled converters, Ac voltage controllers
- 3 Explicate the working and applications DC choppers and cycloconverters
- 4 Analyze and compare the performance of DC and AC machines in various drive applications

Text Books:

1. *Dubey, G.K., Doradla, S.R., Joshi, A. and Sinha, R.M.K., Thyristorised Power Controllers, New Age International (P) Limited, Publishers (2004).*
2. *Rashid, M., Power Electronics, Prentice Hall of India (2006).*
3. *Bimbhra, P.S., Power Electronics, Khanna Publishers (2012).*
4. *N. Mohan, T. M. Undeland, W.P Robbins, —Power Electronics, Converters, Applications & Design, Wiley India Pvt. Ltd.*
5. *Daniel.W.Hart, "Power Electronics", Mc GrawHill Publications 2010.*
6. *Joseph Vithayathil, —Power Electronics, Tata McGraw Hill.*
7. *Simon Ang, Alejandro Oliva, "Power-Switching Converters" Taylor and Francis group*
8. *Krishnan, R., Electric Motor Drives: Modeling, Analysis, and Control. Prentice Hall, (2001).*

Reference Books:

- 1 *Mohan, N., Underland, T. and Robbins, W. P., Power Electronics: Converter Applications and Design, John Wiley (2007) 3rd ed.*
- 2 *Bose, B.K., Handbook of Power Electronics, IEEE Publications*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessionals (Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30